Grand Capstone Defense Presentation

Adaptive Electrical Energy Forecasting, Anomaly Detection & Concept Drift Monitoring

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Reference Architecture: Im et al. (2024), *Frontiers in Big Data* (7:1308236)

Slide 1: Executive Overview & The Problem of Non-Stationarity

- **The Problem**: In real-world electrical grids and residential meters, telemetry is highly dynamic:
- Transient spikes from major appliances
- Sensor dropouts and missing gaps
- Structural concept drift (occupancy shifts, seasonal changes)
- **Failure of Static Models**: Traditional static baselines (SARIMA or fixed deep neural nets) degrade severely because they cannot adapt to non-stationary shifts.
- **Our Solution**:
- Distributed event-driven streaming with Apache Kafka
- Rolling 1-hour aggregation (`energy.hourly`) with watermark lateness handling
- Multi-objective Dynamic Model Selection (SARIMA vs. LSTM vs. Lightweight)
- Non-parametric Rolling MAD Anomaly Detection decoupled from ADWIN Concept Drift Monitoring
- Autonomous online model fine-tuning and hot activation

Slide 2: Streaming Architecture & Decoupled Intelligence

- **Unified Event Envelope**:
- `trigger_event_id`, `trigger_replay_time_ms`, `emit_time_ms` tracked across every topic for strict latency tracing.
- **The Core Decoupling Rule**:

$$\text{Point Anomaly (Spike/Dropout)} \neq \text{Concept Drift (Distribution Shift)}$$

- Point anomalies (surges/inverter trips) are alerted on `energy.anomalies` but isolated from model training.
- Concept drift is monitored on prediction residuals $e_t = |y_t - \hat{y}_t|$ via ADWIN.
- **Storage & Visualization Tier**:
- Real-time metrics scraped via Telegraf into InfluxDB 2.7.
- Live monitoring dashboards provisioned in Grafana 11.3.

Slide 3: Controlled A/B/C Benchmark Experimental Findings

- **Dataset**: Evaluated on all 7,918 hours of the 2010 test split under identical chronological conditions.
- **Configuration Comparison**:
- **Config A (Fixed SARIMA Baseline)**: Overall MAE = 0.2752 kW $\rightarrow$ Post-Drift MAE = 0.5105 kW (+85.8% error increase).
- **Config B (Fixed LSTM Baseline)**: Overall MAE = 0.2302 kW $\rightarrow$ Post-Drift MAE = 0.7064 kW (+208.2% error surge).
- **Config C (Full Adaptive Pipeline)**: Overall MAE = **0.2228 kW** $\rightarrow$ Post-Drift MAE = **0.3795 kW**.
- **Key Finding**:
- The Adaptive Pipeline reduces post-drift forecast error by **46.3% vs. Fixed LSTM** and **25.7% vs. Fixed SARIMA**.

Slide 4: Real-Time Detection Efficacy & Adaptation Latency

- **Anomaly Detection Performance**:
- **Recall on Injected Faults**: 100.0% (Flagged all 7 labelled evaluation anomalies; applies specifically to this benchmark set).
- **Precision & F1-Score**: 0.0012 and 0.0024 (reflects high sensitivity across 7,918 stream hours).
- **Detection Latency**: 0.0 steps (instantaneous single-step classification at event arrival).
- **Breakdown Point**: 50% (rolling MAD guarantees extreme spikes never shift baseline).
- **Concept Drift & Adaptation**:
- ADWIN flagged 15 confirmed distribution shifts.
- Triggered 15 online fine-tuning events on historical regime windows with zero future data leakage.
- **Mean Retraining Duration**: **4.20 ms** (maximum 38.28 ms), ensuring zero pipeline lag.

Slide 5: System Telemetry & Operational Production Readiness

- **Throughput**: **169.8 records/second** (well exceeding 60 rec/hr requirement).
- **Inference Latencies**:
- **P50 Latency**: < 1.00 ms
- **P95 Latency**: 1.79 ms
- **P99 Latency**: 7.86 ms
- **Resource Footprint**:
- Memory: 320.58 MiB RSS
- CPU: 98.85% (single-core equivalent, vs. 608% for unconstrained PyTorch).
- **Quality Assurance**:
- **73 / 73 automated unit and integration tests passing (100% green)**.
- Schema Contract v1.0.0 validated across all 7 Kafka topics.